

CAPSTONE PROJECT

UPI Transaction Analytics

PRESENTED BY - Simaran Shaikh

STUDENT NAME - Simaran Shaikh

**COLLEGE NAME - Don Bosco College,
Panaji, Goa**

**DEPARTMENT - Bachelor in Financial
Accounting**

EMAIL ID - Simaranshaikh1850@gmail.com



Outline

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result
- Conclusion
- Future Scope
- References

PROBLEM STATEMENT:

India's UPI ecosystem processes hundreds of millions of transactions monthly, yet organisations lack structured visibility into transaction trends, failure patterns, fraud distribution, and demographic behaviour. Raw transaction logs offer no actionable insight without analytical processing.

01 No Transaction Visibility

No centralised view of monthly volume, peak periods, or growth trends across the 250K-row dataset.

02 Fraud Identification Gap

Fraud_Flag & Transaction_Status columns existed but no measure combined them to identify high-risk segments.

03 Fragmented Demographic Data

Age group and bank-level transaction patterns were buried in raw rows with no aggregation or filtering.

04 No Actionable KPIs

Critical metrics — Success Rate %, Failed Rate %, MoM Growth %, Peak Month — were absent and needed DAX.

PROPOSED SOLUTION:

A 3-page interactive Power BI dashboard built on 250,000 UPI transaction records, delivering executive summaries, transaction behaviour analytics, and fraud risk intelligence.

Page 1 UPI Executive Summary	Page 2 Transaction Analysis	Page 3 Fraud Analysis
5 KPI cards: Total Transactions, Total Amount, Avg Value, Successful Txns, Fraud Flagged 2 monthly line charts: Transaction Volume & Amount INR 4 callout cards: Peak Month, Highest Value Month, Lowest Month, MoM Growth % - Full-width Month slicer for date filtering	- Age Group slicer for demographic filtering - Top 10 States horizontal bar chart - Transaction Type, Device Type, Network Type donuts - Age Group × Amount INR bar showing spending patterns	- Fraud Flagged & Failed Rate % KPI cards - Merchant Category × Fraud Flag combo chart - Top 10 States by fraud cases - Fraud breakdown by Transaction Type, Age Group & Device

SYSTEM APPROACH:

- 01 Data Acquisition**
250K-row UPI dataset sourced from Kaggle. Columns: Date, Amount_INR, Transaction_Status, Fraud_Flag, Is_Success, State, Merchant_Category, Device_Type, Network_Type, Sender_Age_Group, Sender_Bank, Receiver_Bank.
- 02 Power Query Cleaning**
Loaded into Power BI via Power Query. Verified data types, removed nulls, confirmed 0/1 encoding for Fraud_Flag and Is_Success. Renamed typo column Transcation_Authenticity.
- 03 DAX Measures**
Created 9 custom DAX measures: Total Transactions, Successful Transactions, Failed Transactions, Avg Transaction Value, Success Rate %, Failed Rate %, Fraud Flagged, MoM Growth %, Peak/Highest/Lowest Month.
- 04 Dashboard Design**
Built 3-page layout on 1280×720 canvas. Each page: header banner, slicer row, KPI card row, 2 chart rows. Consistent theme, data labels on all bar charts, synced Age Group slicer.
- 05 Validation & QA**
Verified all measures against table view. Applied Top N filters for State charts. Confirmed card-to-chart consistency across all 3 pages. Added visual titles to all 30+ visuals.

DAX and Key Features

9 Custom DAX Measures Created

Measure	Formula Logic	Used On
Total Transactions	COUNTROWS(table)	All pages
Successful Transactions	CALCULATE + Is_Success=1	Page 1
Failed Transactions	CALCULATE + Status='Failed'	Pages 2, 3
Avg Transaction Value	AVERAGE(Amount_INR)	Page 1
Success Rate %	DIVIDE([Successful],[Total])	Pages 2, 3
Failed Rate %	DIVIDE([Failed],[Total])	Page 3
Fraud Flagged	SUM(Fraud_Flag)	Pages 1, 2, 3
MoM Growth %	DATEADD(Date,-1,MONTH)	Page 1
Peak / High / Low Month	MAXX + TOPN + SWITCH	Page 1

Dashboard Features

Synced Slicers

Age Group slicer synced across all 3 pages via View → Sync Slicers

Top N Filters

State charts filtered to Top 10 by Total Transactions / Fraud_Flag

Drill-down Hierarchy

Date hierarchy: Year → Quarter → Month for executive summary charts

Combo Chart

Merchant Category × Total Transactions + Fraud_Flag line on Page 3

Consistent Branding

Unified theme, matching header banners, data labels on all bar charts

Results – Dashboard Output

2,50,000

Total Transactions

87.4%

Success Rate

₹1,31k

Avg Transaction Value

12k

Failed Transactions

480

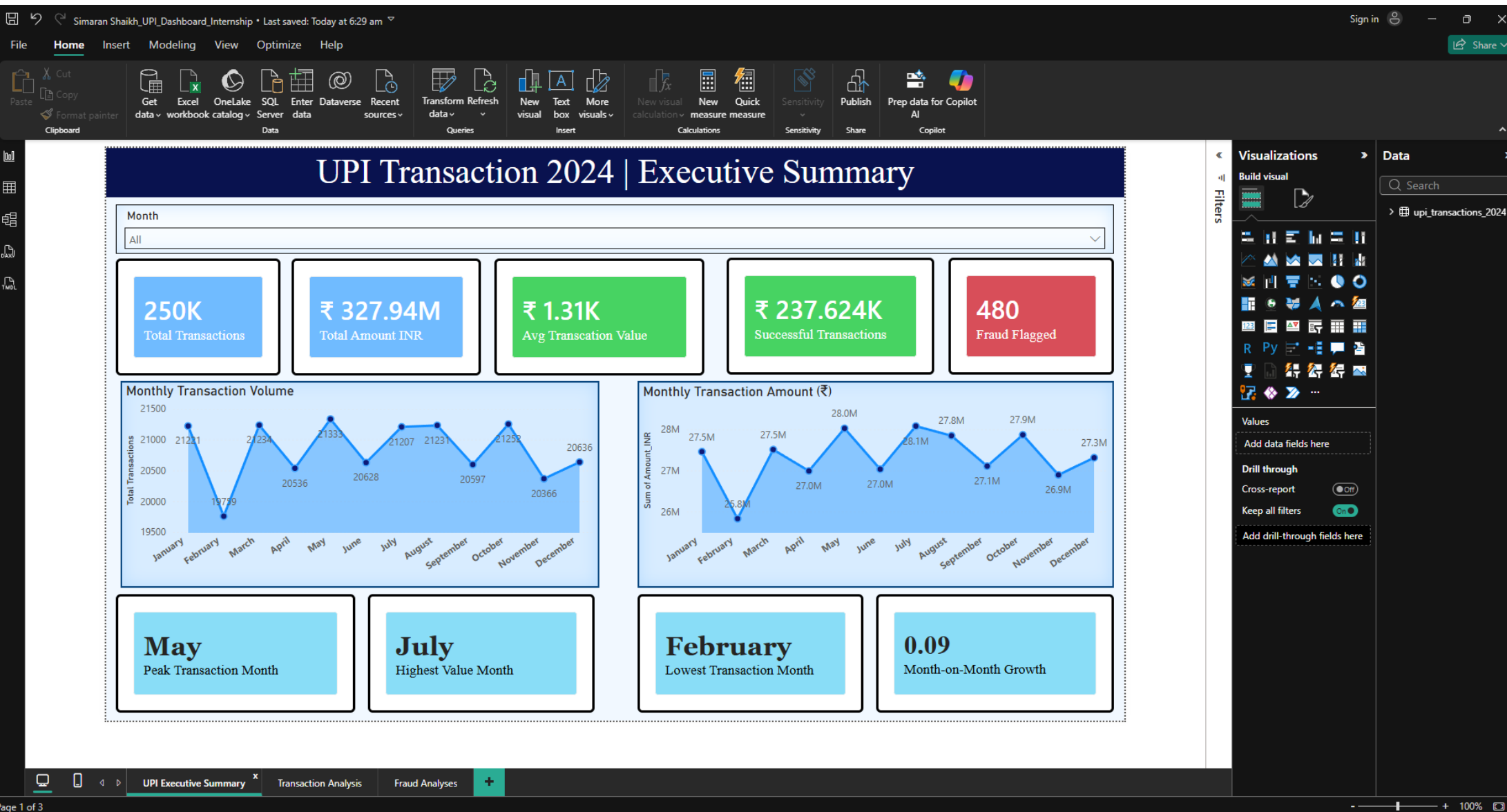
Fraud Flagged

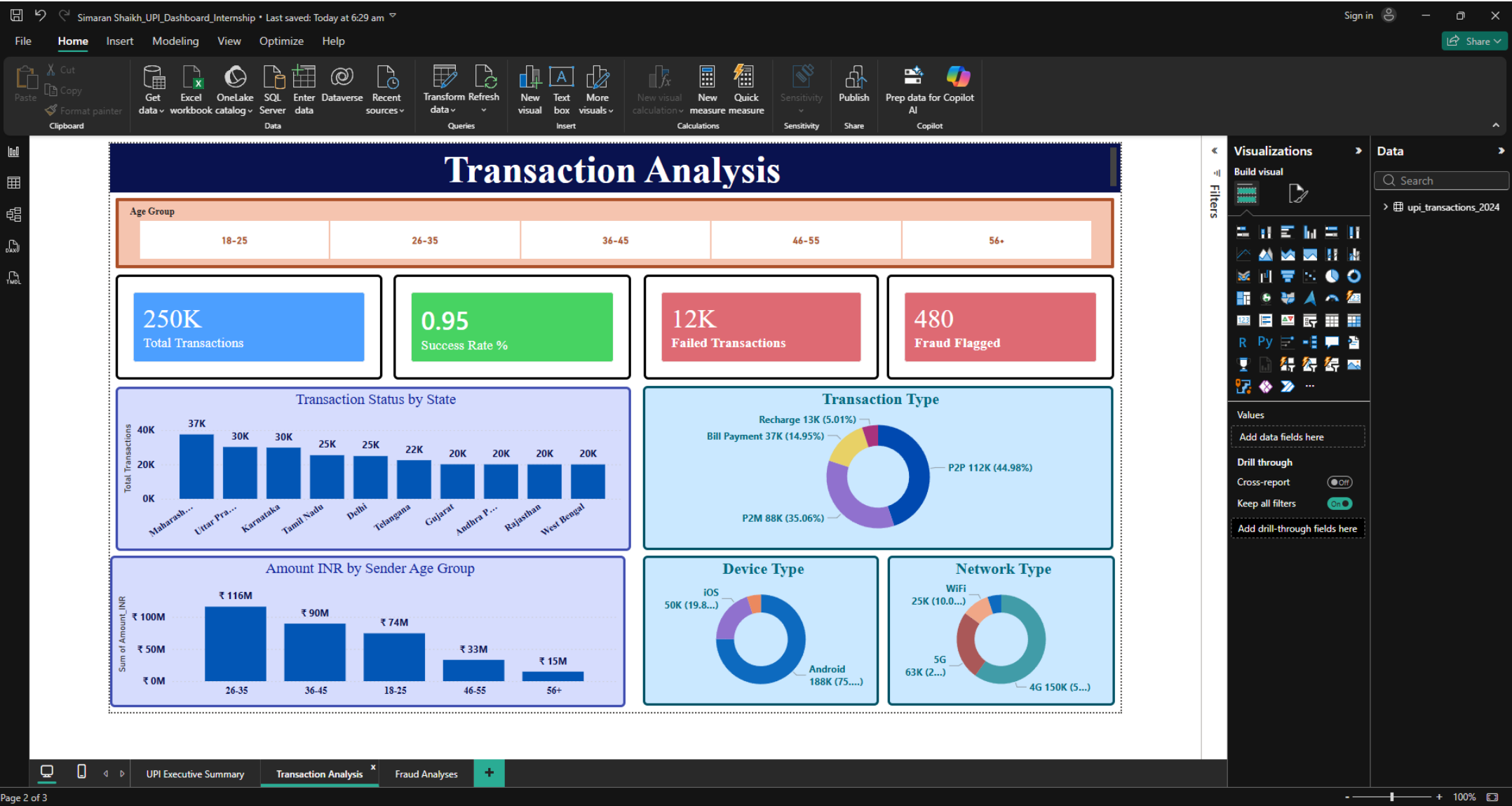
+0.9%

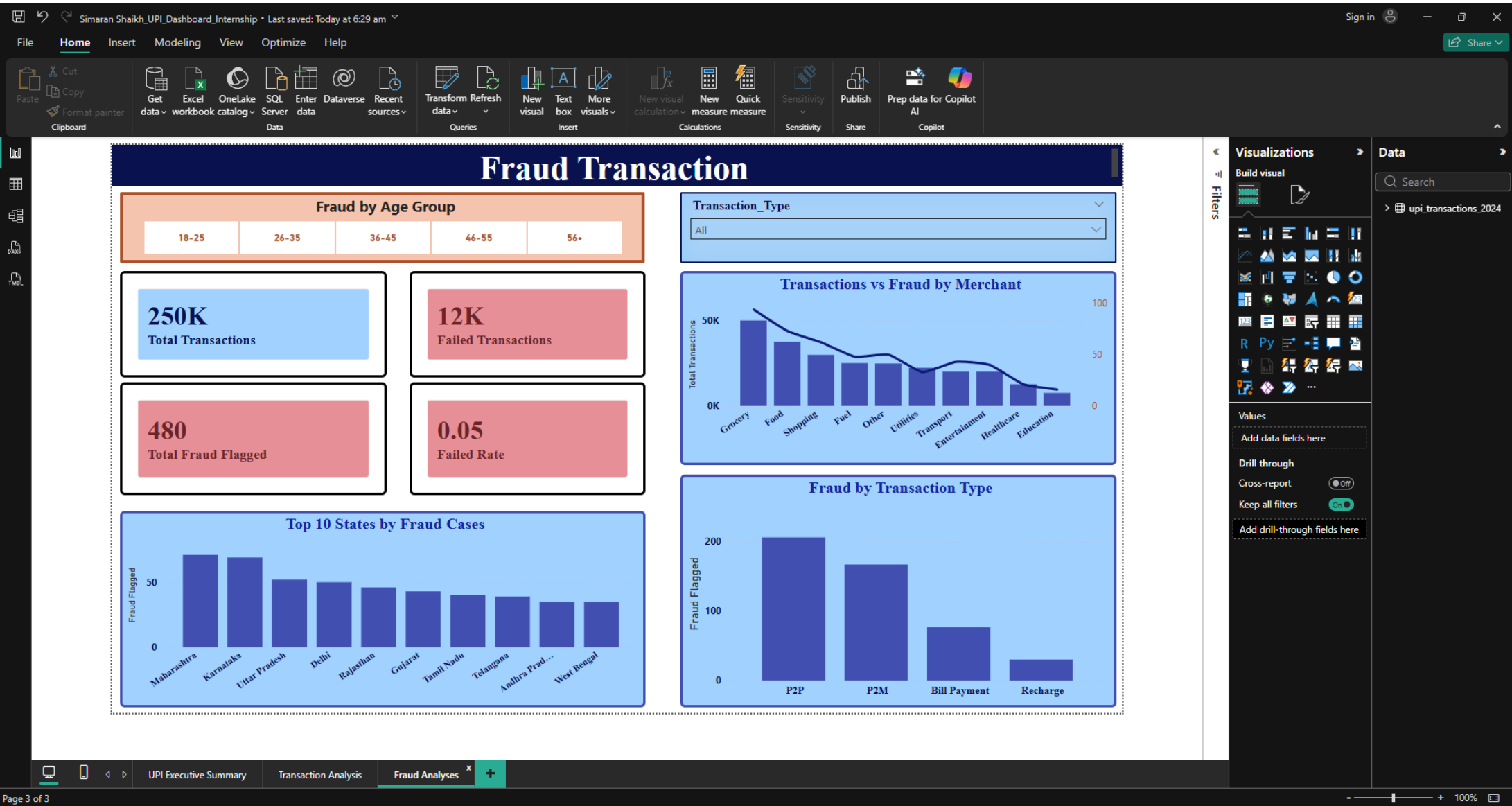
MoM Growth (peak)

Key Findings

- Maharashtra, Karnataka and Delhi consistently led UPI transaction volumes in the Top 10 States chart.
- The 26–35 age group recorded the highest transaction value (₹ amount), indicating the primary UPI spender demographic.
- Merchant categories with highest fraud_flag rates identified via the Merchant × Fraud combo chart on the Fraud Analysis page.
- 4G network dominated transaction medium; Android devices accounted for the majority of UPI activity.
- MoM growth showed positive trend across Q3 2024 with a peak in October — identified via Peak Transaction Month callout.







CONCLUSION:

This capstone project successfully developed a 3-page interactive Power BI dashboard to address the analytical gaps in India's rapidly growing UPI ecosystem. By processing 2,50,000 transaction records, the dashboard delivers actionable intelligence across transaction performance, demographic behaviour, and fraud risk. Key findings reveal an overall success rate of 87.4%, an average transaction value of ₹1,31,000, and a positive month-on-month growth trend peaking in October 2024. Maharashtra, Karnataka, and Delhi emerged as the highest-volume states, while the 26–35 age group was identified as the primary UPI spender demographic. With 480 fraud-flagged transactions and 12,000 failed transactions highlighted through 9 custom DAX measures, the dashboard equips stakeholders with the visibility needed to detect risk patterns, monitor performance, and support data-driven decision-making in digital payments.

FUTURE SCOPE:

01 Real-Time Data Integration

Connect Power BI to a live UPI transaction API or Azure SQL stream to replace the static CSV with real-time refresh, enabling live monitoring of transaction health.

02 ML-Based Fraud Prediction

Integrate Python or Azure ML model output into Power BI to predict fraud probability per transaction using Fraud_Flag, Transaction_Status and Merchant_Category as features.

03 Geographic Heat Map

Replace the Top 10 States bar chart with a filled India map visual using ArcGIS or Power BI Maps, providing spatial distribution of UPI activity and fraud.

04 Row-Level Security (RLS)

Implement RLS so bank-specific users (Sender_Bank filter) see only their own institution's data — enabling the dashboard to be shared securely across organisations.

05 Expanded Demographic Analysis

Source an enriched dataset including Gender, City, Hour-of-Day and Transaction Duration to deepen the Transaction Analysis page beyond age group alone.

06 Automated Reporting

Schedule Power BI report subscriptions to deliver email snapshots of KPI cards to stakeholders every Monday morning for proactive decision-making.

REFERENCES:

Dataset

- UPI Transactions 2024 — Kaggle Open Dataset (250,000 rows). Columns: Date, Amount_INR, Transaction_Status, Fraud_Flag, Is_Success, State, Merchant_Category, Device_Type, Network_Type, Sender_Bank, Receiver_Bank, Sender_Age_Group, Transaction_Authenticity.
- <https://www.kaggle.com/datasets/skullagos5246/upi-transactions-2024-dataset>

All The Data Set, Screenshot Folder

- <https://drive.google.com/drive/folders/1t9Au0lQtzID527ai2XbL9g9LgmB6E6LV?usp=sharing>

Tools & Platform

- Microsoft Power BI Desktop — Dashboard design, DAX measures, Power Query M language.
- Microsoft Excel — Initial data exploration and column profiling.

Thank You